Team Contribution Report

Project: University Library Analytics Data Warehouse
Course: Data Mining and Data Warehousing
Date: October 31, 2026

---

1. Team Members

- **Ferdinand** (Data Architect & Database Designer)
- **NYIRANEZA PHELOMENE** (ETL Specialist & Data Quality Lead)
- **Bineza Jean Paulin** (Analytics, Security & Reporting Lead)

2. Individual Contributions

Ferdinand (Data Architect)

- Designed the overall Data Warehouse architecture, choosing the Bottom-Up (Kimball) approach.
- Designed the Star Schema dimensional model, including the `library_usage_fact` table and all dimension tables (`dim_date`, `dim_student`, `dim_resource`, `dim_room`, `dim_time`).
- Wrote the complete SQL DDL scripts to create the database schemas, primary keys, and foreign key constraints.
- Handled the implementation of the MySQL database environment.

NYIRANEZA PHELOMENE (ETL Specialist)

- Developed the ETL pipeline using Python, Pandas, and SQLAlchemy.
- Extracted data from the three heterogeneous sources (CSV, Excel, MySQL).
- Implemented data cleansing rules: standardized department names (e.g., "CS" to "Computer Science"), fixed mixed date formats to ISO 8601, and removed over 50 duplicate records across the datasets.
- Handled missing data appropriately (e.g., flagging unreturned books and replacing missing student IDs with "UNKNOWN").
- Authored the Data Quality Report documenting the before-and-after metrics.

Bineza Jean Paulin (Analytics & Security Lead)

- Developed the 10+ SQL queries for business intelligence, including the 5 required OLAP operations (Drill-down, Roll-up, Slicing, Dicing, and Pivot analysis).
- Built the interactive Streamlit dashboard featuring the Executive, Departmental, and Operational views.
- Implemented Role-Based Access Control (RBAC) in the application to restrict data access based on user roles (Admin, Analyst, Department Head).
- Authored the Security Policy and Backup/Recovery documentation.

3. Challenges Faced

- **Inconsistent Date Formats:** The digital usage dataset contained three completely different date formats (`01/15/2024`, `2024-15-01`, `Jan 15, 2024`). This was challenging to parse until we utilized Pandas' `format='mixed'` function to standardize everything to `YYYY-MM-DD`.
- **Handling Duplicates and Nulls:** Deciding how to handle missing `ReturnDate` values (books not yet returned) required careful thought so that we didn't delete valid transactions. We solved this by keeping the NULLs and calculating loan days only for returned books.
- **Database Schema Conflicts:** During the ETL process, Pandas would sometimes overwrite the staging table schema. We resolved this by explicitly writing `CREATE TABLE` SQL statements and using Pandas `to_sql(..., if_exists='append')`.
- **GitHub Version Control:** Managing the folder structure and resolving merge conflicts when uploading the final files was a steep learning curve for the team.

4. Lessons Learned

- **Data Quality is 80% of the Work:** We learned that building the database is easy, but cleaning the messy legacy data is the most time-consuming and critical part of a Data Warehouse project.
- **The Value of Dimensional Modeling:** Understanding how to structure a Star Schema made us realize why it is the industry standard—it makes writing complex analytical SQL queries much simpler and faster.
- **Interactivity Matters:** Moving from static Jupyter Notebook outputs to an interactive Streamlit dashboard showed us how data can actually be used by non-technical stakeholders (like the Library Director) to make decisions.
- **Security Cannot Be an Afterthought:** Implementing RBAC taught us that protecting sensitive student data is a strict requirement in any data system.